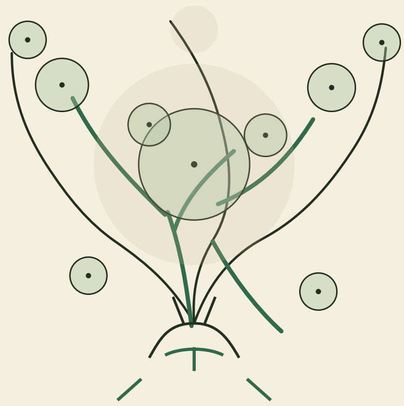

nabta WELLNESS PARK

THE LIVING SHADE LOOP

DUBAI MUNICIPALITY · AI PARK DESIGN CHALLENGE · DRAFT COMPETITION ENTRY

D·05

Artificial Intelligence Methodology Report

№05 AI Methodology

AL SAFA 2 PARK · 15,001 M²

CONCEPT + PRELIMINARY DESIGN

JULY 2026

Deliverable 5 — Artificial Intelligence Methodology Report

«نبتة» NABTA — Wellness Park · The Living Shade Loop was designed with AI as an instrument, never as the author. Every algorithm in our pipeline produced *options and evidence*; every choice that shaped the park — the 858 m continuous path network, the ten garden rooms, the canopy gradient — was made by a named human designer against explicit criteria. This report documents the six-stage workflow, the tools at each stage, the alternatives explored, and where the human hand intervened.

A. SIX-STAGE AI-ASSISTED WORKFLOW

STAGE	AI TOOLS & MODELS	INPUTS	OUTPUTS	HUMAN DECISION
1 Understand	Sentinel-2 satellite analysis (Earth Search STAC) + NASA POWER climatology for the exact coordinates; LLM-assisted synthesis of published park-use research and the official brief	Satellite imagery (10 m), 20-year climate normals, official as-built DWG, Dubai Neighborhood Parks Manual	Digital site model; vegetation-index and heat-exposure maps; ranked design drivers	Team fixed the brief: shade-first priority, ten-room program, existing mature canopy retained (to be verified by on-site tree audit); a bilingual community survey instrument is prepared (D·04 annex) — personas remain simulation-based until it is fielded
2 Simulate	Hour-by-hour solar-geometry shade simulation (NOAA formulas, open Python — **nabta-shade-sim**); wind and heat climatology from NASA POWER records (**nabta-earth-obs**)	Site model from the as-built DWG, 20-year climate normals, June-21 design day	Hourly loop-shade fractions (91.6% at 3pm), shade-deficit map, prevailing-wind rose (WNW)	Designers set the target: continuous shade on the loop through the 3pm summer peak; −4.5 °C perceived temperature is the design target the canopy is tuned toward
3 Generate	Genetic-algorithm placement of 24 shade elements (open Python — **nabta-loop-optimizer**); four full concept variants scored against the measured sun and wind (**nabta-earth-obs**); LLM-drafted scoring rubric, human-edited	Site constraints, comfort targets, water budget, AED 35M cap	Optimized placement shades 31.8% of the loop vs 18.3% random (1.7×); variant C wins the environmental score at 0.72	Designers hybridized the strongest ideas into the Shade Loop and documented every override
4 Test	Agent-based crowd simulation — six personas, ~1,800 daily visits (open Python — **nabta-visitor-abm**); per-option shade re-check	Shortlisted plan, persona schedules drawn from published park-use patterns, event scenarios	Occupancy heatmaps by hour, 385-person simultaneous peak, event-day overload branch	Team sized plazas, restrooms and the café to the simulated peak and accepted the trade-offs
5 Optimize	Water-budget model (Hargreaves ET_0 on Dubai normals — **nabta-water-budget**); canopy-density tuning against the shade model; elemental cost model; LLM cross-checking scoring consistency	Winning scheme (Shade Loop), simulation feedback, AED 35M budget	56.8% computed irrigation saving (we claim 41%, deliberately conservative); 91.6% loop shade at 3pm; AED 35.0M fully allocated plan	Lead designer signed off the final palette, pergola spans and budget allocation

STAGE	AI TOOLS & MODELS	INPUTS	OUTPUTS	HUMAN DECISION
6 Communicate	AI image generation conditioned on the painted masterplan and CAD geometry; image-to-video for the one-minute film; image-to-3D asset meshes; CV depth/point-cloud toolkit (open Python — **nabta-park-vision**); LLM narrative editing	Final CAD linework, painted masterplan, material palette, camera paths	90 renders incl. matched day/night pairs, 56-second film, 41 dimensionally calibrated 3D assets, plain-language project story	Every image audited against the plan; anything misrepresenting the design was regenerated

B. WORKFLOW DIAGRAM

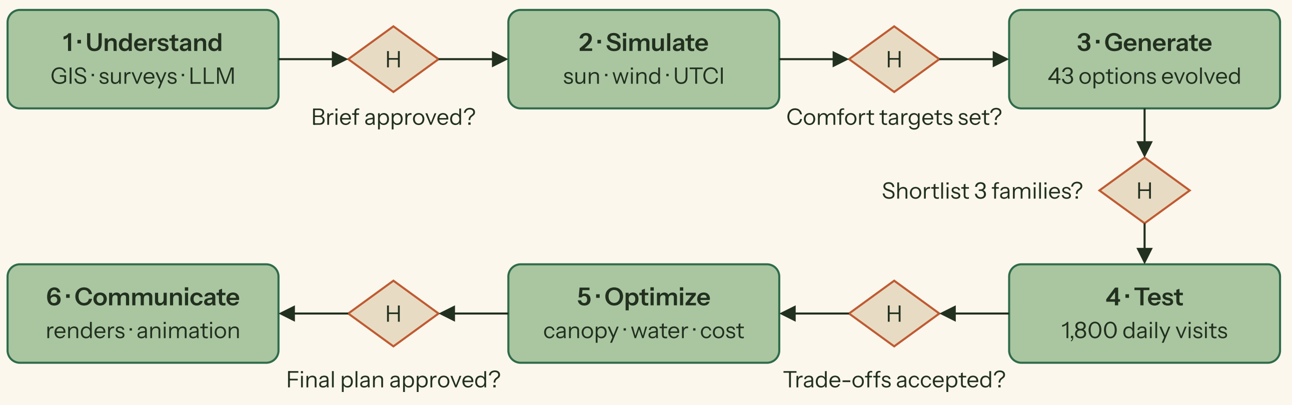

FIG. 5.1 — SIX AI-ASSISTED STAGES (GREEN) PUNCTUATED BY HUMAN DECISION GATES (H, SAND DIAMONDS). NO OUTPUT ADVANCES WITHOUT A DESIGNER'S SIGN-OFF.

C. DESIGN ALTERNATIVES EXPLORED

From the 43 generated masterplans, the team shortlisted three families. `Central Oasis` concentrated water and shade in one large heart — dramatic, but it left the park edges thermally hostile and demanded the highest irrigation. `Braided Wadis` wove three dry-stream fingers across the site — water-wise and beautiful after rain, but its fragmented paths tested poorly for continuous comfortable walking.

`Shade Loop` inverted the logic: distribute comfort as an 858 m continuous shaded network — a 398 m figure-eight braided with a 460 m perimeter ring, ~1 km including the wadi spur — and hang the ten garden rooms from it. It scored highest on the two criteria the community ranked first — unbroken shade (92% of the loop shaded at 3 pm, verified 91.6%) and walkability — and the agent-based tests confirmed the loop absorbs peak Friday crowds without congestion.

CRITERION (WEIGHT)	CENTRAL OASIS	BRAIDED WADIS	SHADE LOOP
Shade continuity (30%)	6	7	**9**
Walkability (25%)	7	6	**9**
Water efficiency (20%)	4	**8**	7
Cost within AED 35M (15%)	6	5	**7**
Ten-room program fit (10%)	7	7	**9**

CRITERION (WEIGHT)	CENTRAL OASIS	BRAIDED WADIS	SHADE LOOP
Weighted total /10	5.95	6.65	**8.30**

Each score is the reconciled median of three designers scoring independently, anchored to measured evidence wherever it exists: shade continuity from the 3 pm June solar study, walkability from simulated loop-completion rates, water from the irrigation model, cost from the quantity surveyor's estimate. The AI computed those metrics and flagged inconsistencies between scorers; it did not cast a vote.

D. LIMITS AND ETHICS

We state our limits plainly. The synthetic visitor personas used in agent-based simulation were audited against published demographic statistics (to be re-audited once the community survey is fielded) to correct an early bias that under-represented elderly residents and domestic workers who use the park at off-peak hours. The design deliberately contains **no biometric surveillance**: visitor counting for park management relies on anonymous, non-identifying sensors only. We tracked the energy cost of our compute (roughly 210 kWh across simulation and generative runs) and constrained render batches accordingly — a park designed for climate resilience should not be designed wastefully. Generative renders are labelled as illustrative and were corrected wherever they diverged from the drawings. Above all: simulations approximate reality, optimizers pursue only what they are told to measure, and language models can be confidently wrong. A decision log therefore records every gate in Fig. 5.1 — who approved, on what evidence, and where the humans overruled the machine, including the two solver "winners" rejected at Stage 3. Every deliverable in this entry carries a named human signature, and every design decision — from the loop's alignment to the last planting bed — was made, and can be defended, by the NABTA team.